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PERFECT SCORE OF 150/150 AS A TESTAMENT TO EXECEPTIONAL E-LEARNING METHODS

UPES UNIVERSITY OF TOMORROW

Elements of AIML

Unit 05 – Lecture 01: AI for Society, Women & Environment

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Topic focus: **responsible AI** — fairness, privacy, safety, sustainability

Repository: `https://github.com/tali7c/Elements-of-AIML`

Sources: NIST, *AI Risk Management Framework (AI RMF 1.0)*, 2023.;Barocas, Hardt, and Narayanan, *Fairness and Machine Learning* (online book), selected sections.;Knaflic, *Storytelling with Data*, selected chapters on communicating results.

Quick Links

Why Bias/Fairness Privacy Environment Summary

Agenda

Why responsible AI?

Bias and fairness

Privacy and security

Environment and sustainability

Summary

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- Discuss environmental and sustainability considerations for AI.

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- Sustainability: energy/compute cost considerations

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- Where does **human oversight** enter?

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- **Human role**: borderline cases should be reviewed, not auto-rejected blindly.

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- **Proxy features**: non-sensitive variables correlate with sensitive ones

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Fairness can mean different things depending on context:

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- Error rates should be similar across groups (group fairness).
- Outcomes should not systematically disadvantage protected groups.

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Fairness Metrics (Examples, not exhaustive)

- **Demographic parity**: positive prediction rates similar across groups

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- **Error-rate balance**: similar FPR/FNR across groups

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- Models can leak information (membership inference, memorization).
- Sharing data across teams/vendors increases risk.

Mitigations: data minimization, access control, anonymization (with caution), audits.

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- Monitoring and human oversight are often required.

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Before deployment, ask:

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- How will drift be detected and handled?

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- Are results consistent across groups/subpopulations?
- What data/PII is used and who has access?
- How will drift be detected and handled?
- Is there a human escalation/appeal path?

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- Choose "good enough" accuracy when marginal gains cost huge resources.

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- Evaluate outcomes across groups; investigate gaps.
- Use participatory design: stakeholders help define success and risks.

Summary

- Responsible AI requires fairness, privacy, safety, and accountability checks.
- Bias can enter via data, labels, and historical decisions.
- Sustainability matters: compute/energy trade-offs should be considered.

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Exit Questions

1. List three sources of bias in ML datasets.
2. Give one example where fairness concerns are critical.
3. What is one privacy risk of ML models (not just datasets)?
4. Name two ways to reduce environmental impact of ML systems.